

TEACHER ANSWER KEY: Propagation & Reflection of Light

Course: Grade 10 Physics

Topic: Light Propagation, Reflection, & Mirrors

Part 1: Multiple Choice (1 point each)

1. (b) In straight lines

Explanation: The principle of rectilinear propagation states that light travels in straight lines in a homogeneous transparent medium.

2. (a) Converging beam

3. (c) They reflect light towards our eyes

4. (d) Virtual and of the same size

5. (c) The Sun

Explanation: The Sun is a primary source that emits its own light. The Moon, mirrors, and paper only reflect light.

6. (b) Diffuse reflection

Part 2: Conceptual Questions & Calculations

7. The Laws of Reflection (2 points):

- **First Law:** The incident ray, the reflected ray, and the normal at the point of incidence all lie in the same plane.
- **Second Law:** The angle of incidence ($\hat{i}$) is equal to the angle of reflection ($\hat{r}$). ($\hat{i} = \hat{r}$)

8. Types of Reflection / The Puddle Effect (3 points):

When the puddle is perfectly calm, the water surface is smooth, causing **regular (specular) reflection** where parallel rays reflect in the same direction, creating a clear image. When it rains, the surface becomes rough and bumpy. This causes **diffuse reflection**, where parallel light rays scatter and reflect in different directions, causing the image to appear distorted or unclear.

9. Spectrum & Color (2 points):

White light contains all colors of the visible spectrum. The leaf appears green because it **reflects** the green portion of the spectrum into our eyes, while **absorbing** the rest of the colors.

10. Reflection Calculation (3 points):

Working/Logic: A clock face is a full circle (360°) divided into 12 hours. Therefore, the angle between each hour mark is $360^\circ/12 = 30^\circ$.

- a) **Total angle:** From 12 o'clock to 5 o'clock is 5 hours.
 $5 \times 30^\circ = \mathbf{150^\circ}$.
- b) **Angle of incidence ($\hat{i}$):** The normal line bisects the total angle perfectly. Since $\hat{i} + \hat{r} = 150^\circ$ and $\hat{i} = \hat{r}$:
 $2\hat{i} = 150^\circ \implies \hat{i} = \mathbf{75^\circ}$.
- c) **Angle of reflection ($\hat{r}$):** By the second law of reflection ($\hat{i} = \hat{r}$):
 $\hat{r} = \mathbf{75^\circ}$.

Part 3: Ray Diagrams & Plane Mirrors

11. Drawing Reflection (3 points):

Grading Rubric for Student Drawing:

- 1 pt** Straight line for mirror with shading/hashes on the back to denote the non-reflective side.
- 1 pt** Correctly drawn and labeled **Incident Ray** (arrow pointing toward mirror) and **Reflected Ray** (arrow pointing away from mirror).
- 1 pt** Correctly drawn **Normal** (dashed line perpendicular to mirror surface) and correctly identified **Angle of Incidence (i)** and **Angle of Reflection (r)** strictly between the normal and their respective rays.

12. Image Characteristics (4 points):

Award 1 point for any 4 of the following characteristics:

- It is **virtual** (cannot be collected on a screen / appears behind the mirror).
- It is the **same size** as the object ($h = h'$).
- The image distance is **equal** to the object distance from the mirror ($q = p$).
- It is **upright** (not inverted vertically).
- It is **laterally inverted** (left and right are reversed).

End of Answer Key